

TEST - B

Q) IP Parameters:-

i) Find the network ID, First IP, Last IP, Broadcast IP, Prefix length & subnet mask.

ii) 18.18.18.251 $\rightarrow$ class A

Network ID $\rightarrow$ 18.0.0.0

First IP $\rightarrow$ 18.0.0.1

Last IP $\rightarrow$ 18.255.255.254

Broadcast IP $\rightarrow$ 18.255.255.255

Prefix length $\rightarrow$ /8

Subnet mask $\rightarrow$ 255.0.0.0

iii) 172.16.25.31 $\rightarrow$ class B

Network ID $\rightarrow$ 172.16.0.0

First IP $\rightarrow$ 172.16.0.1

Last IP $\rightarrow$ 172.16.255.254

Broadcast IP $\rightarrow$ 172.16.255.255

Prefix length $\rightarrow$ /16

Subnet mask $\rightarrow$ 255.255.0.0

iii) 1.10.10 → class A

Network ID → 1.0.0.0

First IP → 1.0.0.1

Last IP → 1.255.255.254

Broadcast IP → 1.255.255.255

Prefix length → 18

Subnet mask → 255.0.0.0

iv) 50.0.1.0 → class A

Network ID → 50.0.0.0

First IP → 50.0.0.1

Last IP → 50.255.255.254

Broadcast IP → 50.255.255.255

Prefix length → 8

Subnet mask → 255.0.0.0

v) 45.252.31.4 → class A

Network ID → 45.0.0.0

First IP → ~~45.252~~ 45.0.0.1

Last IP → 45.255.255.254

Broadcast IP → 45.255.255.255

Prefix length → 8

Subnet mask → 255.0.0.0

vi) $\overset{N}{72}.\overset{H}{16}.\overset{H}{255}.\overset{H}{31} \rightarrow \underline{\text{class A}}$

Network IP $\rightarrow 72.0.0.0$

First IP $\rightarrow 72.0.0.1$

Last IP $\rightarrow 72.255.255.254$

Broadcast IP $\rightarrow 72.255.255.255$

Prefix length $\rightarrow 8$

Subnet mask $\rightarrow 255.0.0.0$

vii) $128.16.0.10 \rightarrow \underline{\text{class B}}$

Network IP $\rightarrow 128.16.0.0$

First IP $\rightarrow 128.16.0.1$

Last IP $\rightarrow 128.16.255.254$

Broadcast IP $\rightarrow 128.16.255.255$

Prefix length $\rightarrow 16$

Subnet mask $\rightarrow 255.255.0.0$

viii) $180.0.1.0 \rightarrow \underline{\text{Class B}}$

Network IP $\rightarrow 180.0.0.0$

First IP $\rightarrow 180.0.0.1$

Last IP $\rightarrow 180.0.255.254$

Broadcast IP $\rightarrow 180.0.255.255$

Prefix length $\rightarrow 16$

Subnet mask $\rightarrow 255.255.0.0$

*) 193.65.20.1 $\rightarrow$ Class C

Network ID $\rightarrow$ 193.65.20.0

First IP $\rightarrow$ 193.65.20.1

Last IP $\rightarrow$ 193.65.20.254

Broadcast IP $\rightarrow$ 193.65.20.255

Prefix length $\rightarrow$ 24

Subnet mask $\rightarrow$ 255.255.255.0

*) 223.0.0.1 $\rightarrow$ Class C

Network ID $\rightarrow$ 223.0.0.0

First IP $\rightarrow$ 223.0.0.1

Last IP $\rightarrow$ 223.0.0.254

Broadcast IP $\rightarrow$ 223.0.0.255

Prefix length $\rightarrow$ 24

Subnet mask $\rightarrow$ 255.255.255.0

*) 210.210.0.36 $\rightarrow$ Class C

Network ID $\rightarrow$ 210.210.0.0

First IP $\rightarrow$ 210.210.0.1

Last IP $\rightarrow$ 210.210.0.254

Broadcast IP $\rightarrow$ 210.210.0.255

Prefix length $\rightarrow$ 24

Subnet mask $\rightarrow$ 255.255.255.0

*) 169.253.0.10 $\rightarrow$ class B

network ID $\rightarrow$ 169.253.0.0

First IP $\rightarrow$ 169.253.0.1

Last IP $\rightarrow$ 169.253.255.254

Broadcast IP $\rightarrow$ 169.253.255.255

Prefix length $\rightarrow$ 16

Subnet mask $\rightarrow$ 255.255.0.0

*) 99.88.25.142 $\rightarrow$ class A

network ID $\rightarrow$ 99.0.0.0

First IP $\rightarrow$ 99.0.0.1

Last IP $\rightarrow$ 99.255.255.254

Broadcast IP $\rightarrow$ 99.255.255.255

Prefix length $\rightarrow$ 8

Subnet mask $\rightarrow$ 255.0.0.0

* 63.201.63.9 $\rightarrow$ class A

network ID $\rightarrow$ 63.⁰~~201~~.0.0

First IP $\rightarrow$ 63.0.0.1

Last IP $\rightarrow$ 63.255.255.254

Broadcast IP $\rightarrow$ 63.255.255.255

Prefix length $\rightarrow$ 8

Subnet mask $\rightarrow$ 255.0.0.0

*) 23.53.121.40 $\rightarrow$ class A

Network IP $\rightarrow$ 23.0.0.0

First IP $\rightarrow$ 23.0.0.1

Last IP $\rightarrow$ 23.255.255.254

Broadcast IP $\rightarrow$ 23.255.255.255

Prefix length $\rightarrow$ 8

Subnet mask $\rightarrow$ 255.0.0.0

*) 14.0.0.10 $\rightarrow$ class A

Network IP $\rightarrow$ 14.0.0.0

First IP $\rightarrow$ 14.0.0.1

Last IP $\rightarrow$ 14.255.255.254

Broadcast IP $\rightarrow$ 14.255.255.255

Prefix length $\rightarrow$ 8

Subnet mask $\rightarrow$ 255.0.0.0

*) 2.0.0.10 $\rightarrow$ class A

Network IP $\rightarrow$ 2.0.0.0

First IP $\rightarrow$ 2.0.0.1

Last IP $\rightarrow$ 2.255.255.254

Broadcast IP $\rightarrow$ 2.255.255.255

Prefix length $\rightarrow$ 8

Subnet mask $\rightarrow$ 255.0.0.0

* 95.95.0.0 → class A

Network ID → 95.95.0.0

First IP → 95.0.0.1

Last IP → 95.255.255.254

Broadcast IP → 95.255.255.255

Prefix length → 8

Subnet mask → 255.0.0.0

* 220.15.0.136 → class C

Network ID → 220.15.0.0

First IP → 220.15.0.1

Last IP → 220.15.0.254

Broadcast IP → 220.15.0.255

Prefix length → 24

Subnet mask → 255.255.255.0

* 49.0.0.210 → Class A

Network ID → 49.0.0.0

First IP → 49.0.0.1

Last IP → 49.255.255.254

Broadcast IP → 49.255.255.255

Prefix length → 8

Subnet mask → 255.0.0.0